

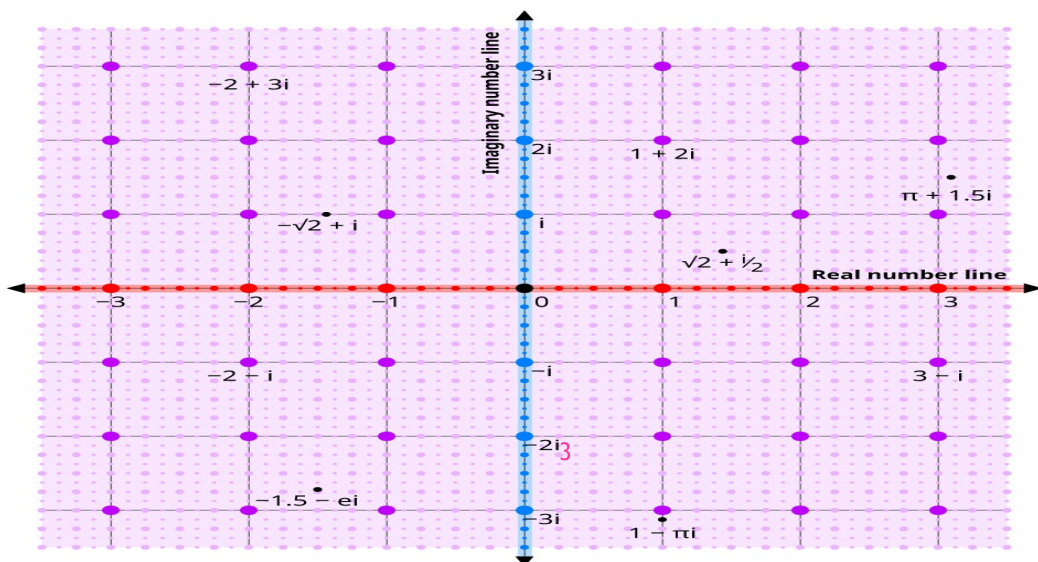
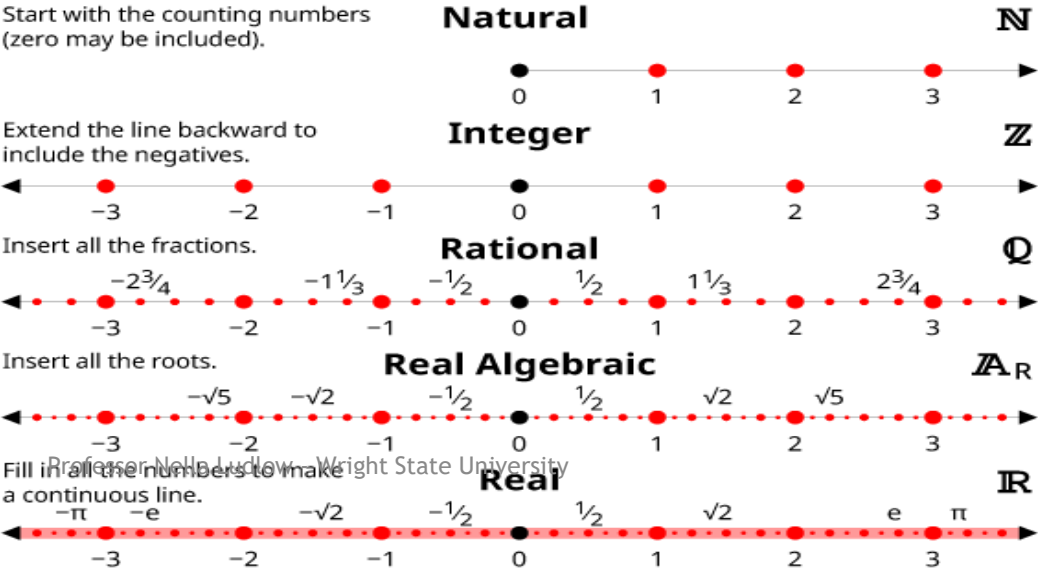
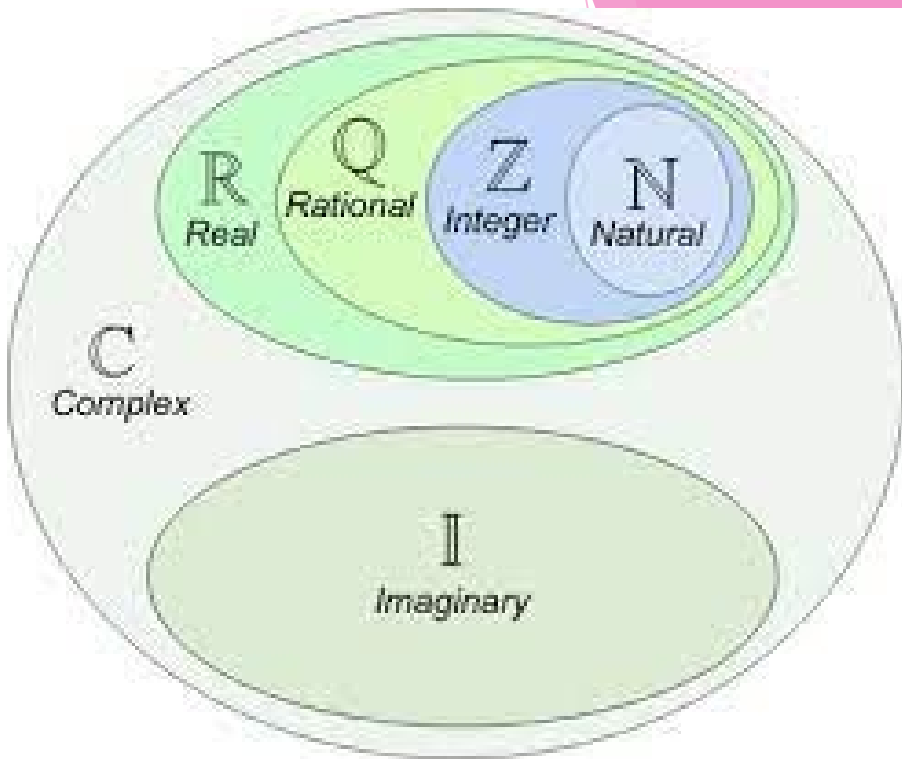
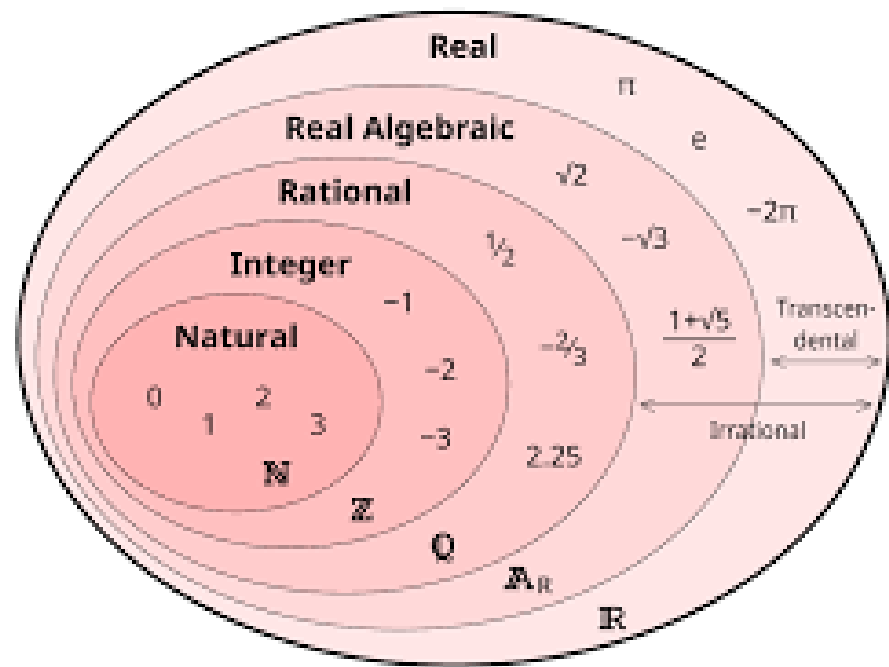
Number Theory, Trigonometry, and Dirac Notation

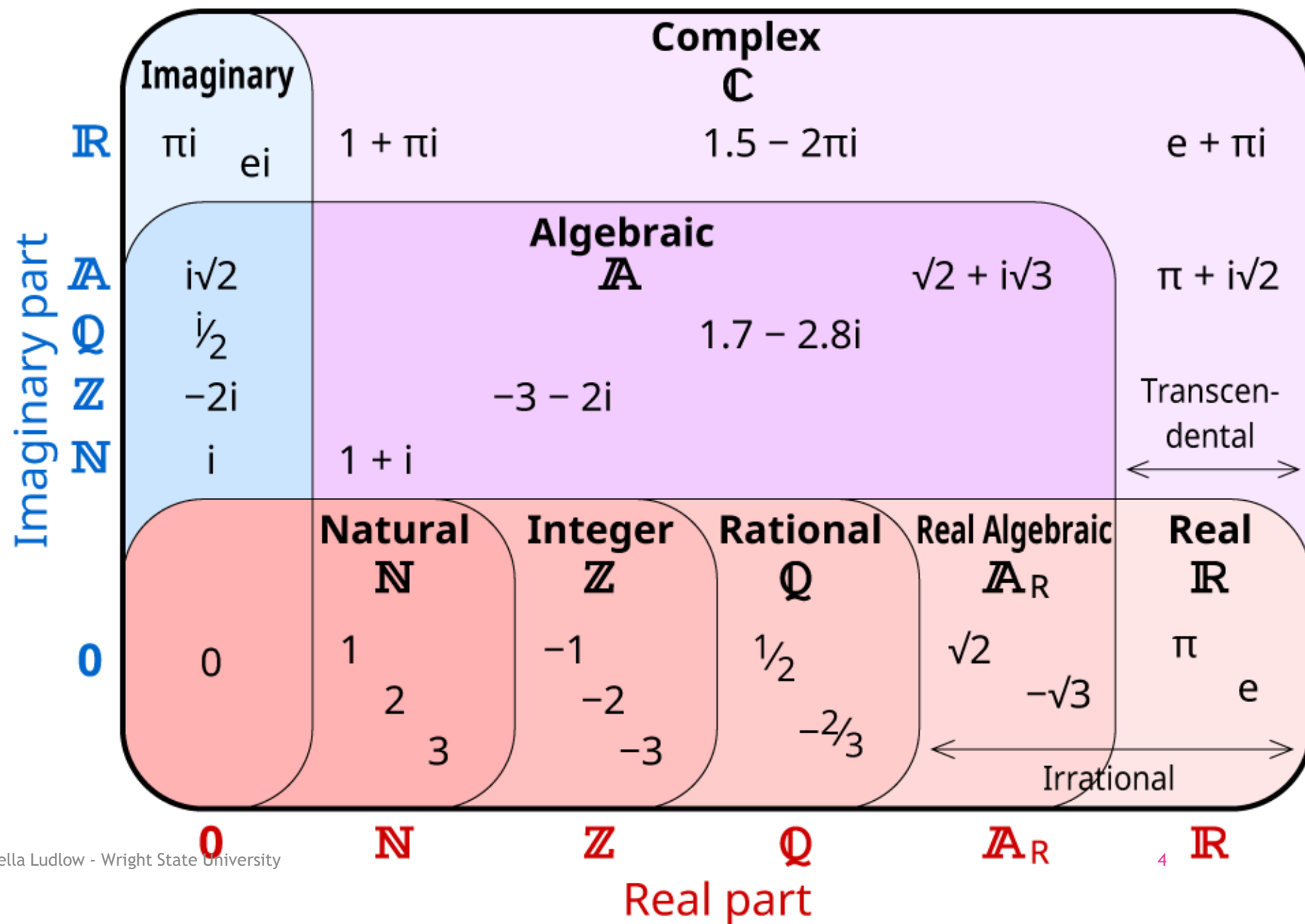
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This Week What You Should Learn

- Quick Review of Quantum Logic, Dirac Notation and Bloch Sphere
- Both 1 qubit and 2-or-more Qubit Logic
- Bloch Sphere Axes
- Why Imaginary Numbers?
- Relationship between Amplitude and Probability

Number Theory





Properties of the Number Sets

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{A}$	$\mathbb{C}$
	Natural	Integer	Rational	Real	Algebraic	Complex
Closed under Addition ¹	•	•	•	•	•	•
Closed under Multiplication ¹	•	•	•	•	•	•
Closed under Subtraction ¹		•	•	•	•	•
Closed under Division ¹			•	•	•	•
Dense ²			•	•	•	•
Complete (Continuous) ³				•		•
Algebraically Closed ⁴					•	•

1. Closed under addition (multiplication, subtraction, division) means the sum (product, difference, quotient) of any two numbers in the set is also in the set.
2. Dense: Between any two numbers there is another number in the set.
3. Continuous with no gaps. Every sequence that keeps getting closer together (Cauchy sequence) will converge to a limit in the set.
4. Every polynomial with coefficients in the set has a root in the set.

The complex numbers are the algebraic completion of the real numbers. This may explain why they appear so often in the laws of nature.

Microsoft Slides: Math Review

Quick review: real and complex numbers

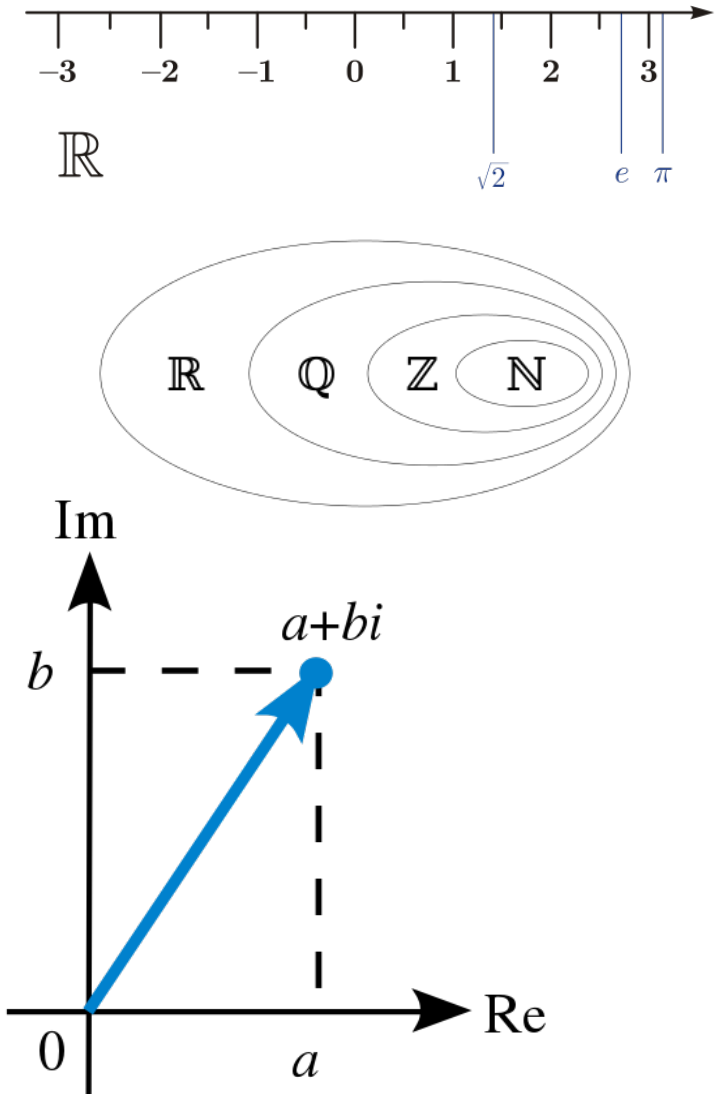
- Real numbers $\mathbb{R}$
 - Can represent distance on a line
 - Includes rational numbers, irrational numbers, natural numbers
- Complex numbers $\mathbb{C}$
 - Expressed as

$$a + bi$$

where $a, b \in \mathbb{R}$, and i is an indeterminate with $i^2 = -1$

- Complex conjugation: $(a + bi)^* = a - bi$
- Squared magnitude:

$$|a + bi|^2 = (a + bi)^*(a + bi) = a^2 + b^2$$



Quick review: polar form

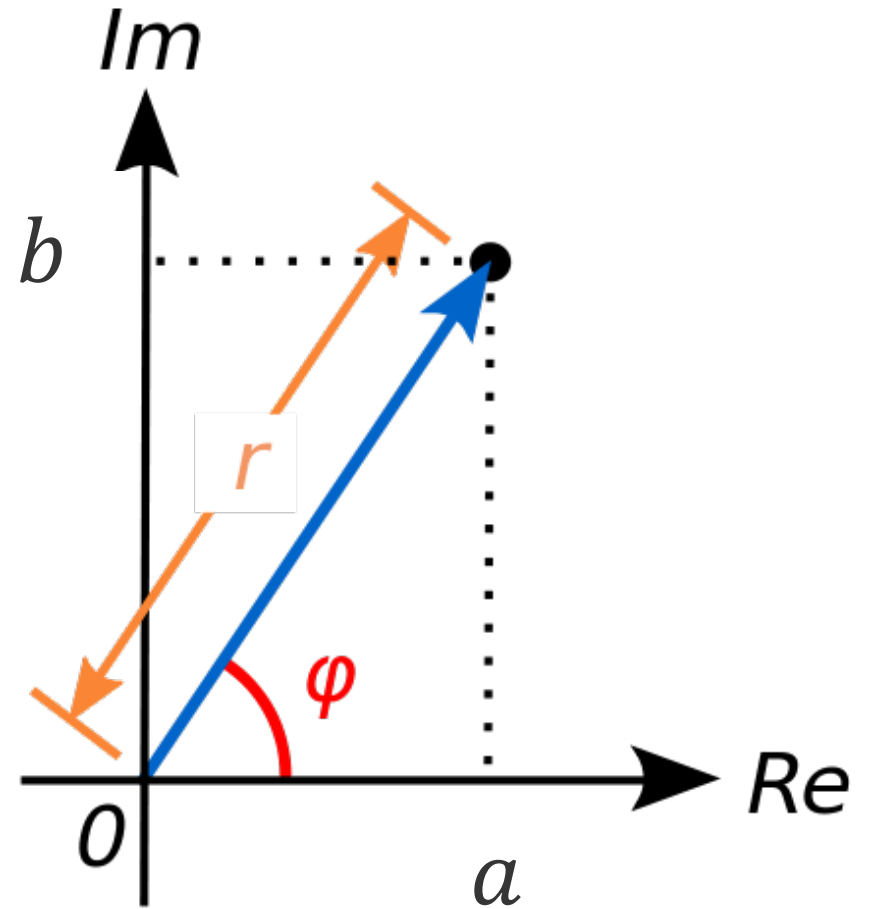
- Let $z = a + bi$

$$r = \|z\| = \sqrt{a^2 + b^2}$$

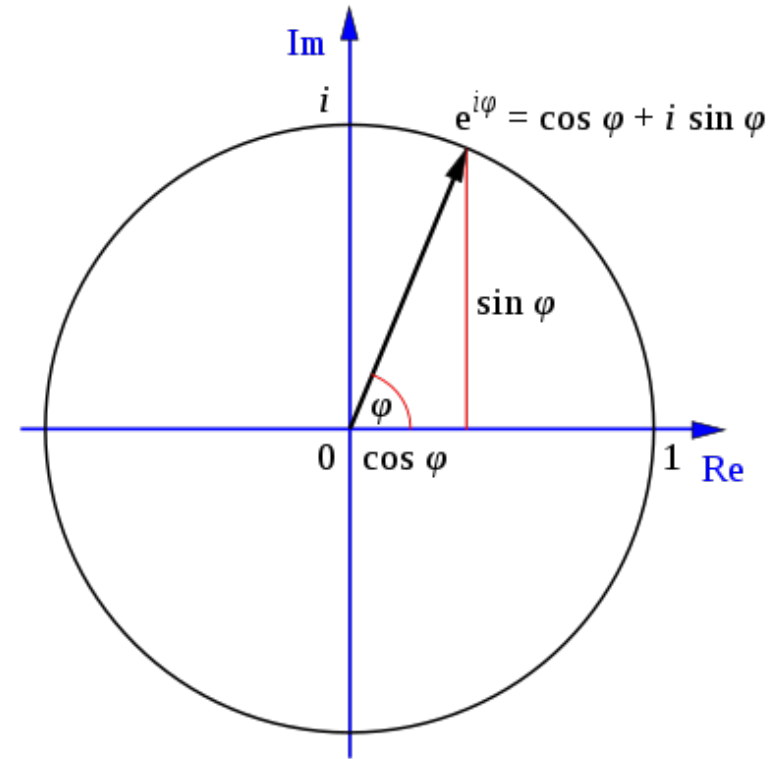
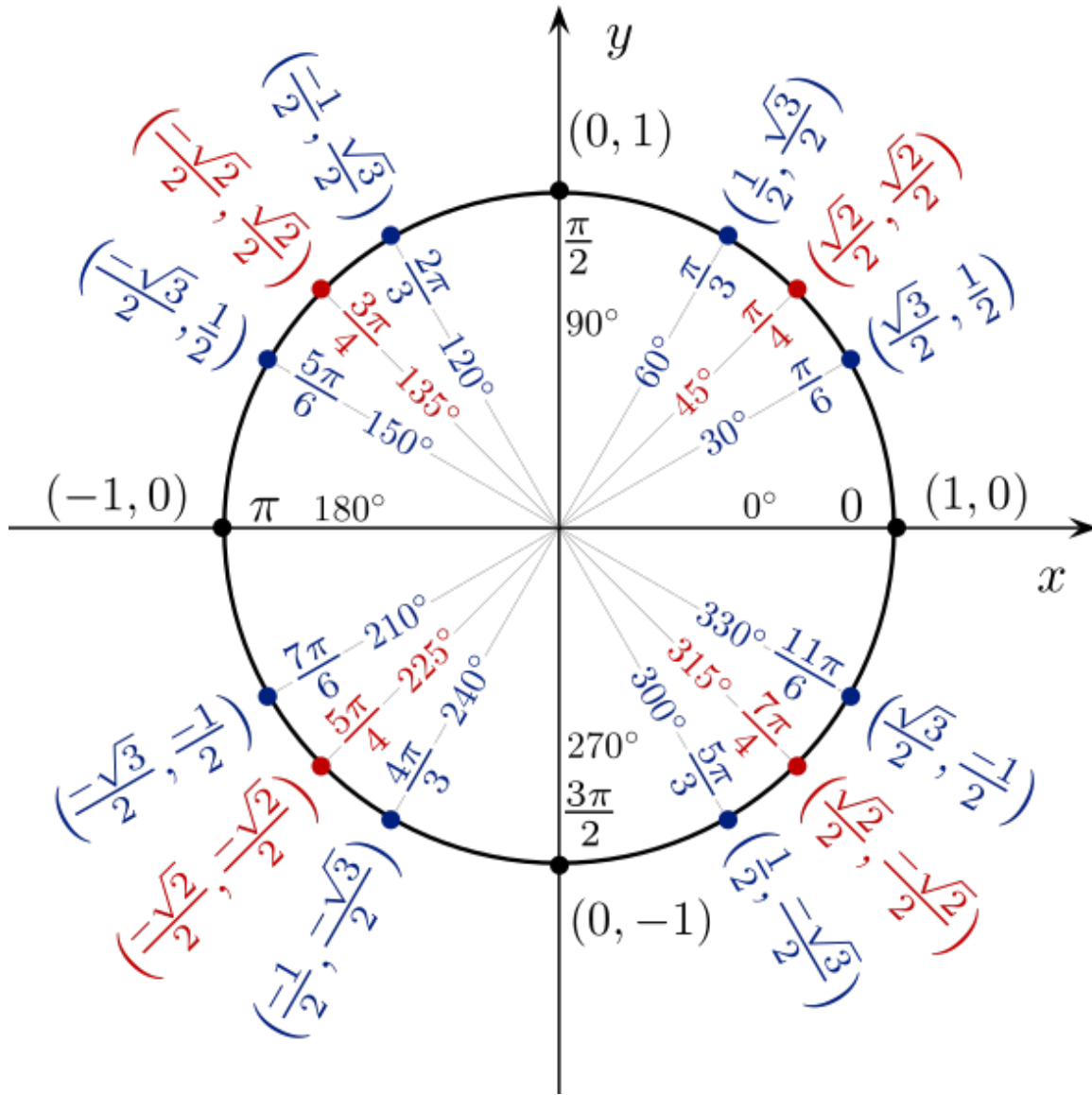
- If $b = 0$, then $r = \|a\|$

- The phase φ is called the *argument*
- (r, φ) specify point in complex plane

$$z = r(\cos\varphi + i\sin\varphi) = re^{i\varphi}$$



Quick review: trigonometric identities



$$\cos^2 \theta + \sin^2 \theta = 1$$

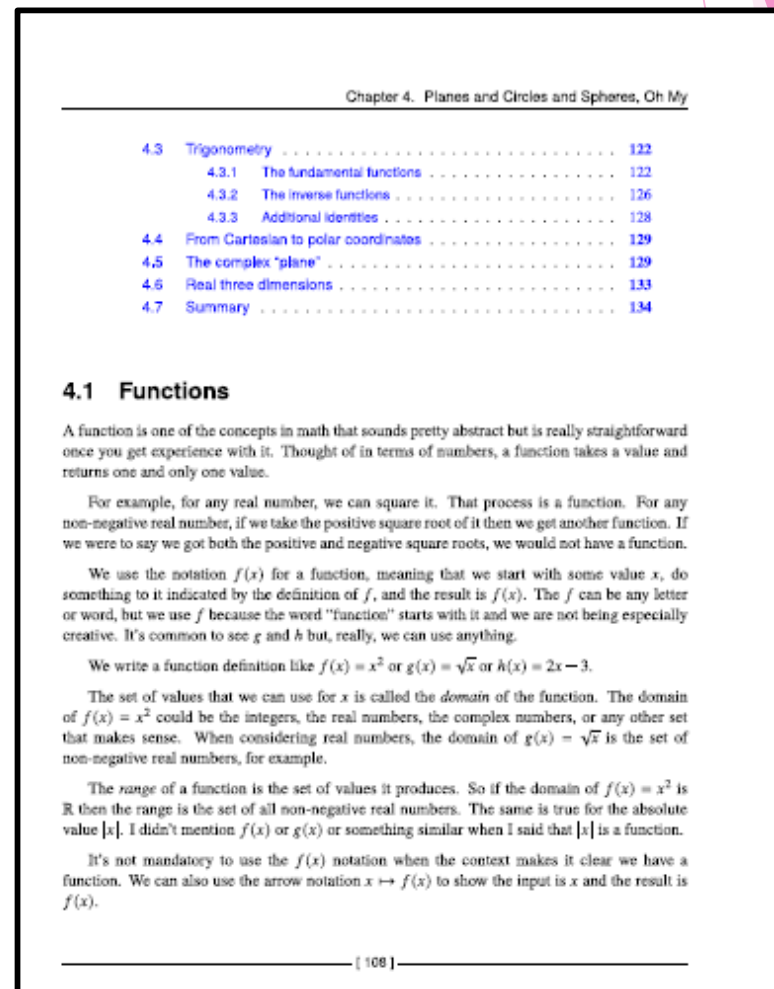
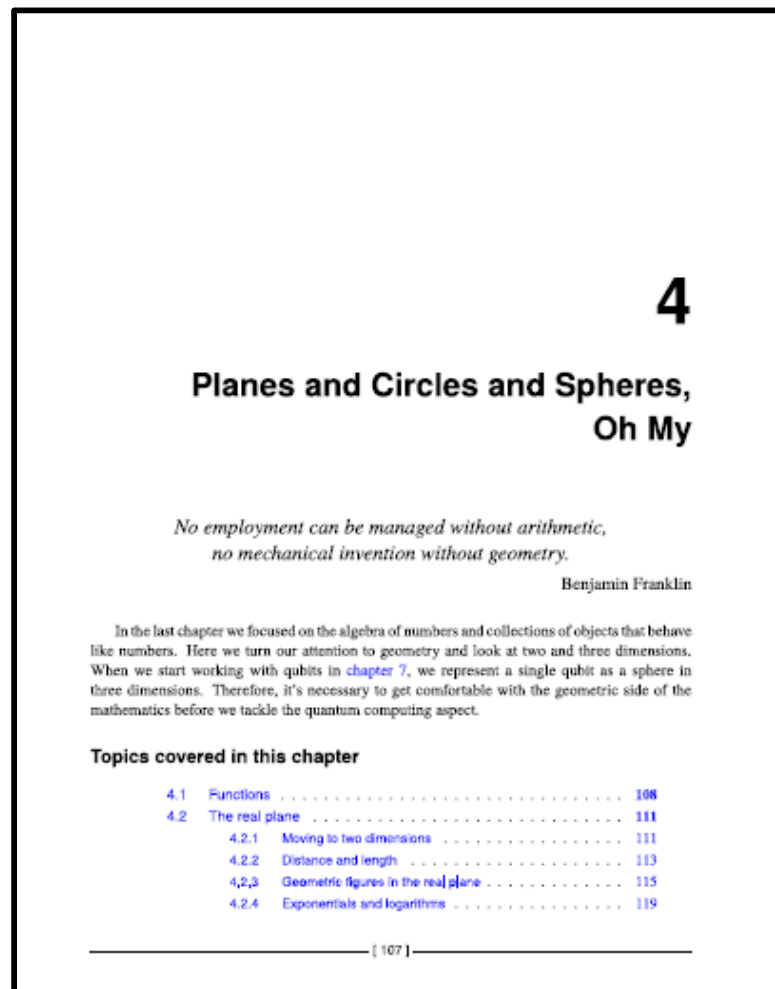
$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{2\pi i} = 1$$

$$e^{\pi i} = -1$$

Need Trigonometry Review

- In Week 3 Optional Reading is:
 - Chapter 4, “Planes Circles and Spheres - Oh My,” Dancing with Qubits,



BLOCH SPHERE

MIT xP

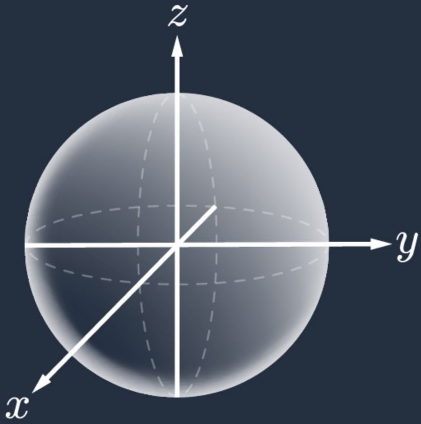
Course Dis

Course > We

< Previous

Bloch Sphere

Three dimensional sphere



Ludlow

Next >

Think of it like the Earth
with 0 at the North Pole
and 1 at the South Pole

There are infinite number
of points on the surface of
the earth



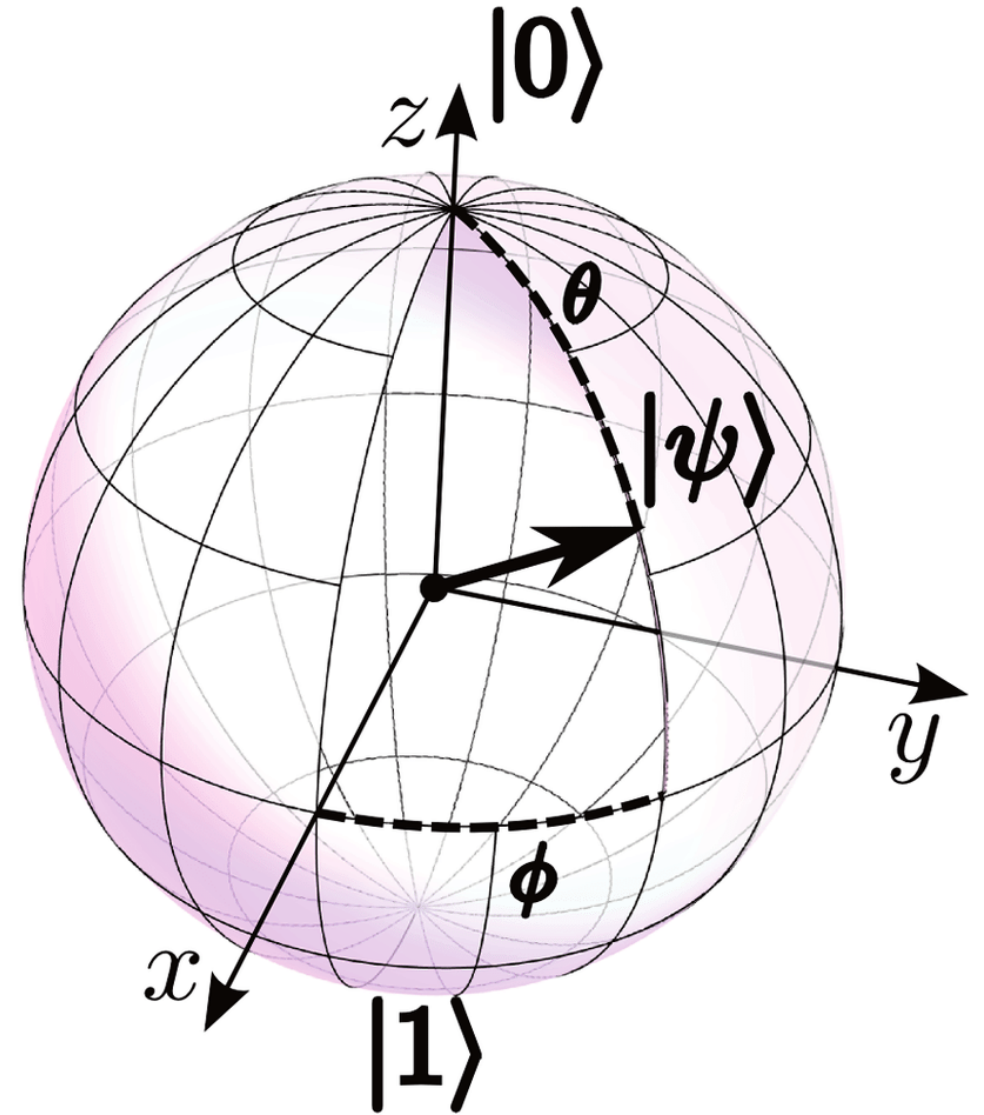
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BLOCH SPHERE

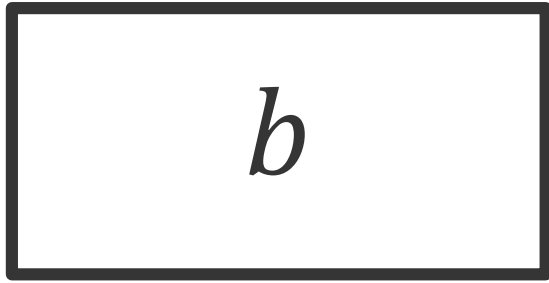
To communicate in mathematical Fields, one must be comfortable with the Greek alphabet.

θ , or theta, is the degrees from positive z axis
 ϕ , or phi, is the angle from positive x axis
 ψ , or psi, is the resultant vector on the sphere



Unit of information

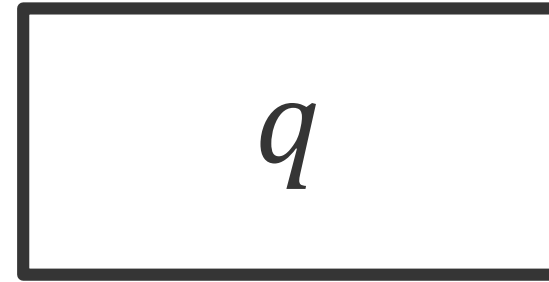
Classical: *bit*



$$b = 0 \text{ or } 1$$

$$b = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{matrix} \text{"0"} \\ \text{"1"} \end{matrix}$$

Quantum: *qubit*



$$q = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} - \begin{matrix} \text{"0"} \\ \text{"1"} \end{matrix}$$

$$\text{with } |c_0|^2 + |c_1|^2 = 1.$$

superposition of 0 and 1 gives quantum advantage

Unit of information

$$q$$

$$q = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

$$\text{with } |c_0|^2 + |c_1|^2 = 1.$$

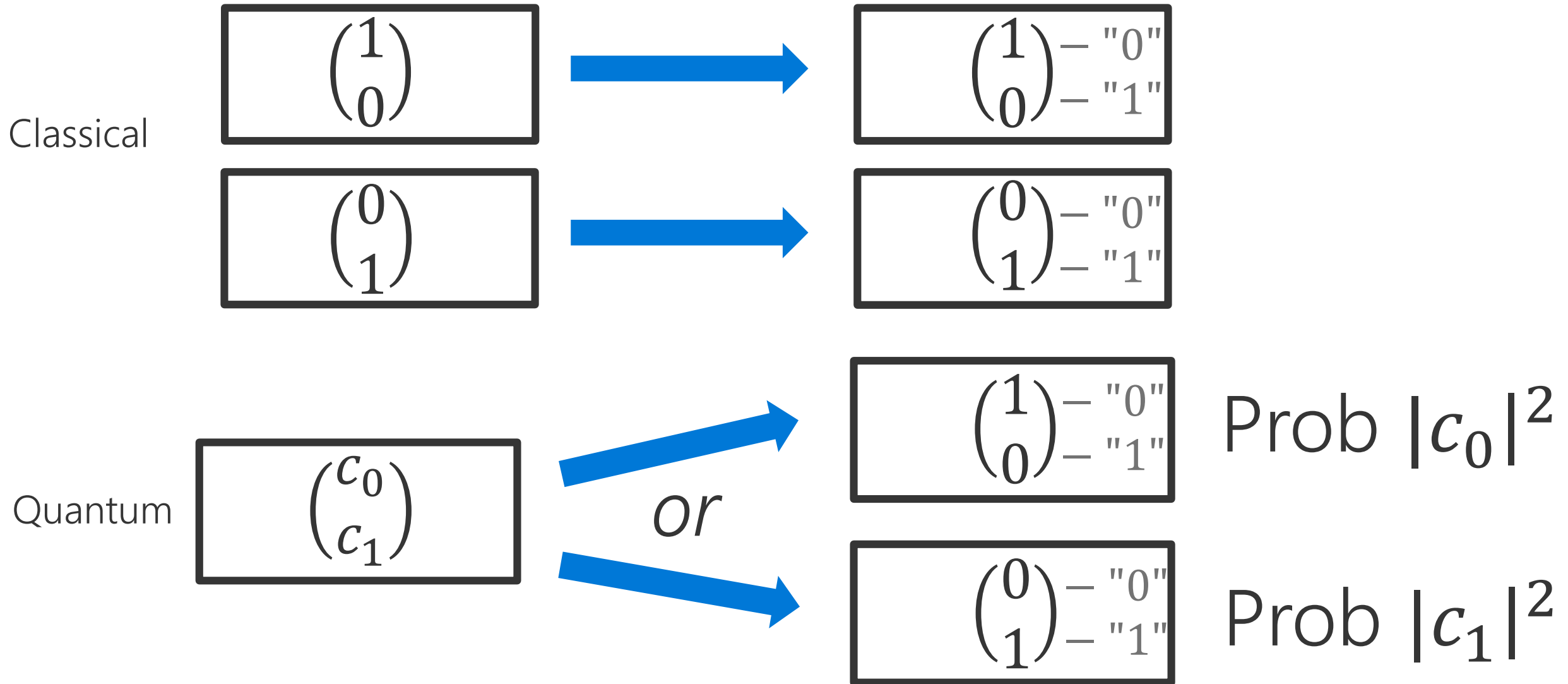
A qubit is in a
linear combination
of basis states

$$q = c_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

Basis states "0" and "1"

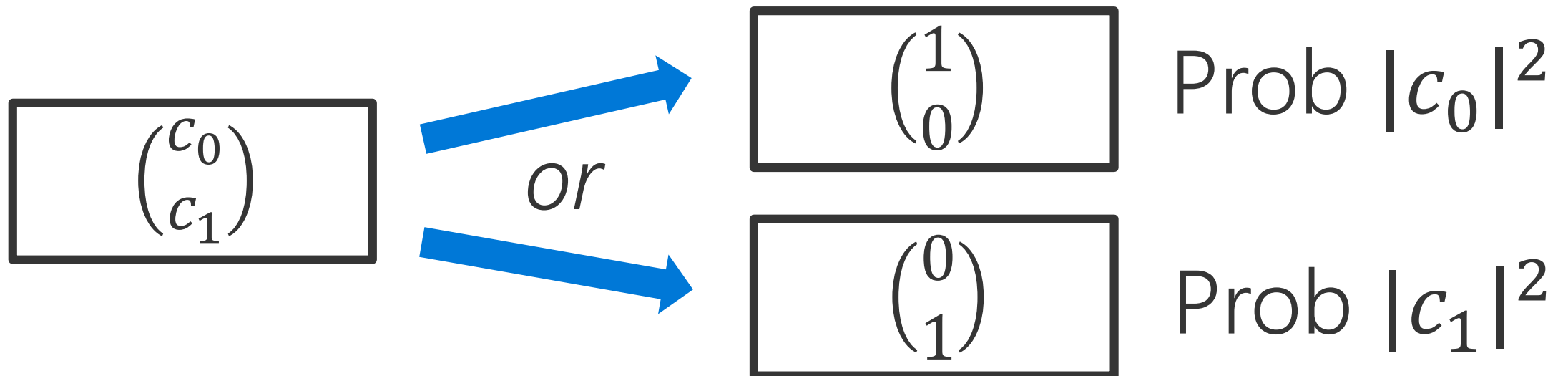
$$c_0, c_1 \text{ are complex: } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \in \mathbb{C}^2$$

Readout: quantum measurement



Readout: quantum measurement

- We need measurement for readout.
- Measurement *restricts power of quantum computing*, preventing us from directly learning c_0 and c_1 .
- Measurement destroys superposition.
- A qubit must be protected from unwanted measurement (known as decoherence).



Example:

$$\begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

This state is in
"equal superposition"

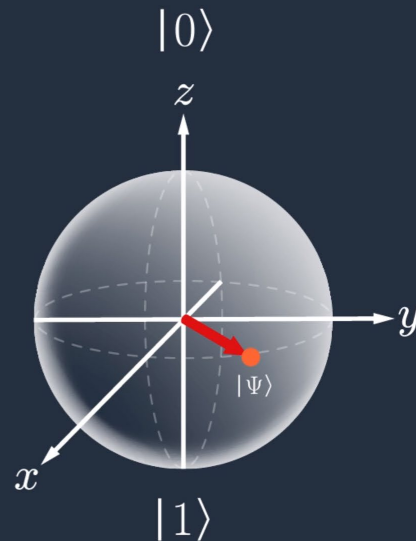
when measured

- yields "0" 50% of the time
- yields "1" 50% of the time

This state is sometimes called the "+" state.

EQUAL SUPERPOSITION

On the
"equator"
of the
Sphere



Equal
Superposition

$$|\Psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

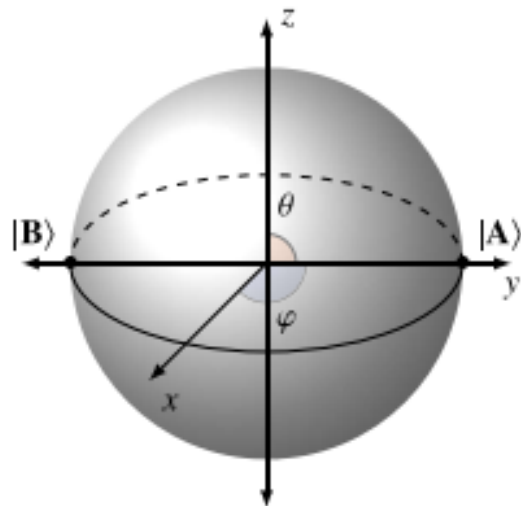
when Bra values $\alpha = \beta$

Bloch Sphere (from our Reading)

The kets $|0\rangle$ and $|1\rangle$ lie on the z axis and $|+\rangle$ and $|-\rangle$ lie on the x axis.

What is the orthonormal basis that would lie on the y axis? Let's call them $|\mathbf{A}\rangle$ and $|\mathbf{B}\rangle$ for now.

The two points in which we are interested have Cartesian coordinates $(0, 1, 0)$ and $(0, -1, 0)$.



We have now seen three pairs of basis elements and they go by different names:

Basis	Common name	Bloch sphere axis name
$\{ 0\rangle, 1\rangle\}$	Computational	Z
$\{ +\rangle, -\rangle\}$	Hadamard	X
$\{ i\rangle, -i\rangle\}$	Circular	Y

The Y basis is the same as the circular basis. In this case we also have the alternative notation $|i\rangle = |\cup\rangle$ and $|-i\rangle = |\cap\rangle$.

NEXT IS 2 QUBITS

